



FACULTY OF ENGINEERING TECHNOLOGY

Subject: comm.& Data Transmission

Mid Exam Fall 22/23

Name:.....

Duration: 50 min

THIS EXAM CONTAINS TWO QUESTIONS

Q1:(Please write your answers into table only, otherwise your answers will be neglected)

Question	1	2	3	4	5	6	7	8	9	10
Answer										
Question	11	12	13	14						
Answer										

- In start stop asynchronous method the synchronization is found at:
a) the byte level b) at the bits level c) at stop level d) at start level
- Analog sinusoidal signal of 10v(p-p) has:
a) An infinite number of values
b) A definite number of values
c) Has only one value
d) Has only two values
- The period of the electrical power in Jordan is equal to :
a) 20ms b) 20 μ s c) 17ms d) 17 μ s
- What is the frequency of AC current which reaches zero value every 0.01 second?
a) 100 Hz b) 50Hz c) 0 d) 500 Hz
- A sinusoidal alternating current having frequency of 50 Hz has a peak value of 10 A. what is the value of current after 1/300 s from zero.
a) 8.66 A b) 181.5 mA c) 50 A d) 16.66 A

6. Two AC currents are given by: $i_1=50\sin(314t+\pi/6)$ and $i_2=20\sin(314t-\pi/4)$
Which of the two current leads the other?
- a) i_1 leads i_2 by 75°
 - b) i_2 leads i_1 by 75°
 - c) i_1 leads i_2 by 15°
 - d) i_2 leads i_1 by 15°
7. Quantization noise in PCM, for a given input, is proportional to the :
- a) sampling rate
 - b) maximum frequency of modulating signal
 - c) number of digits used for coding
 - d) input signal
8. In a communication system, the noise is most likely to affect the signal:
- a) at the transmitter
 - b) at the circuit
 - c) at the channel
 - d) at the destination
9. The human voice signal usually is:
- a) nonperiodic, composite with limited range of frequencies
 - b) periodic, un composite with limited range of frequencies
 - c) nonperiodic, composite with unlimited range of frequencies
 - d) periodic, un composite with unlimited range of frequencies
10. Radio station in FM broadcasting has:
- a) 200 Hz bandwidth
 - b) 200 kHz bandwidth
 - c) 200 MHz bandwidth
 - d) 200 GHz bandwidth
11. An amplifier has an output of 20w.what is the output in dBw:
- a) 13 dBw
 - b) 130 dBw
 - c) 13dBmw
 - d) 130dBmw
12. If we want to transmit 1000 character with each character encoded as 8 bits using asynchronous transmission, then the number of bits need are:
- a) 8000 bits
 - b) 10000 bits
 - c) 4000 bits
 - d) 5000 bits
13. Byte synchronization in synchronous transmission is accomplished by:
- a) physical layer
 - b) data link layer
 - c) application layer
 - d) network layer
14. Forming bits into frames is performed by:
- a) physical layer
 - b) data link layer
 - c) application layer
 - d) network layer

Q2: A power amplifier has an input of 90 mV across 10 K Ω . The output is 7.8 v across 8 Ω speaker:

- a) What is the power gain, in decibel
- b) If the power amplifier is connected to two other amplifiers with gains 10 dB & 20 dB and two filters with attenuations of -2 dB & -4 dB calculate overall gain.

GOOD LUCK

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	A	B	A	A	C	C	A	B
Question	11	12	13	14						
Answer	A	B	B	B						